

3.7 Nitrogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.7 Nitrogen Compounds
Difficulty	Easy

Time allowed: 30

Score: /18

Percentage: /100

Question 1a

Propylamine can be produced from 1-bromopropane.

Complete the equation for the formation of propylamine.

**[2 marks]****Question 1b**

State the necessary reagents and conditions to produce propylamine from 1-bromopropane.

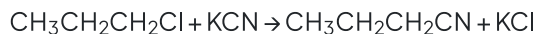
[3 marks]**Question 1c**

Name the reaction mechanism that takes place when propylamine is produced from 1-bromopropane.

[1 mark]

Question 2a

When 1-chloropropane is heated under reflux with ethanolic potassium cyanide, KCN, the following reaction occurs.



i)

Draw the displayed formula of the organic product formed in this reaction.

[1]

ii)

State the IUPAC name of this organic product.

[1]

[2 marks]

Question 2b

Suggest why this reaction is useful to chemists during the synthesis of other organic compounds.

[1 mark]

Question 2c

The $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ can undergo hydrolysis.

i)

Write the equation for the acid hydrolysis of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$.

[2]

ii)

Draw the displayed formula of the intermediate formed when $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ is hydrolysed by sodium hydroxide.

[1]

[3 marks]

Question 3a

This question is about hydroxynitriles.

The hydroxynitrile shown in Fig. 3.1 can be prepared from the reaction between propanal and hydrogen cyanide.

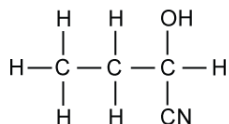


Fig. 3.1

Give the IUPAC name of this hydroxynitrile.

[1 mark]

Question 3b

Using 'curly arrows', complete the reaction mechanism shown in Fig. 3.2 for the reaction between propanal and the cyanide ion, CN^- .

Include any lone pairs of electrons and partial or whole charges.

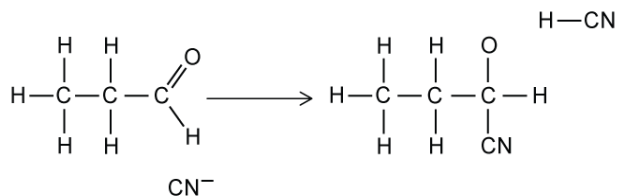


Fig. 3.2

[4 marks]

Question 3c

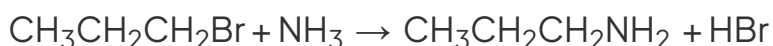
Name the type of reaction mechanism shown in Fig. 3.2.

[1 mark]

Model Answers: Easy

1a

a) The equation for the formation of propylamine is:



- NH_3 ; [1 mark]
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$; [1 mark]

[Total: 2 marks]

- You should know that ammonia is required to form an amine from a halogenoalkane
- If you can't remember, you can work backwards:
 - You should be able to work out the formula of propylamine:
 - As it is an amine, it has the $-\text{NH}_2$ group
 - The propyl- prefix tells you that the NH_2 is attached to a $\text{CH}_3\text{CH}_2\text{CH}_2$ group
 - Its formula is, therefore, $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
 - Add this into the equation:
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \dots \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{HBr}$
 - You can then work out a N and 3 H atoms are required on left-hand side to balance the equation:
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{NH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 + \text{HBr}$

1b

b) The necessary reagents and conditions to produce propylamine from 1-bromopropane are:

- Excess ammonia / NH_3 ; [1 mark]
- Ethanolic / ethanol as a solvent; [1 mark]
- Under pressure; [1 mark]

[Total: 3 marks]

- If the ammonia is not in excess, propylamine can act as a nucleophile and attack 1-bromopropane, forming the secondary amine dipropylamine
- Make sure you know the reagents and conditions required for this reaction

1c

c) The reaction mechanism that takes place when propylamine is produced from 1-bromopropane is:

- Nucleophilic substitution; [1 mark]

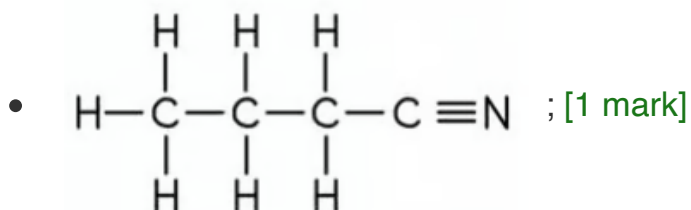
[Total: 1 mark]

- The carbon from the C-Br bond has a partial positive charge
- The lone pair of electrons on the nitrogen atom in ammonia acts as a **nucleophile** and attacks the partially positively charged carbon atom
- The C-Br bond is broken and Br is **substituted** by the amine group
- Therefore, the reaction mechanism is nucleophilic substitution

2a

a)

i) The displayed formula of the organic product formed in this reaction is:



ii) The IUPAC name of this organic product is:

- Butanenitrile; [1 mark]

[Total: 2 marks]

- When halogenoalkanes react with KCN, or NaCN, in ethanol, a nitrile is formed
- Nitriles have the organic functional group -CN
- **Remember:** There is a triple bond between the C and N atoms in the nitrile group

- **Careful:** When naming the organic product, don't forget the C atom in the nitrile group, it is a common mistake for students to overlook this and give the name as propanenitrile

2b

b) This reaction is useful to chemists during the synthesis of other organic compounds because:

- It lengthens the carbon chain; [1 mark]

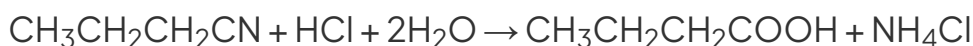
[Total: 1 mark]

- This reaction is of particular importance as it forms a carbon-carbon bond which can lengthen the carbon chain
- This is useful as new compounds can be synthesised which have more carbon atoms
- Sometimes, an organic compound may not be easily available as a raw material, but another organic compound with one less carbon atom may be readily available
 - If this compound with one less carbon atom is made into a halogenoalkane and then refluxed with ethanolic KCN, then a nitrile can be made with the desired number of carbon atoms which can undergo further synthesis

2c

c)

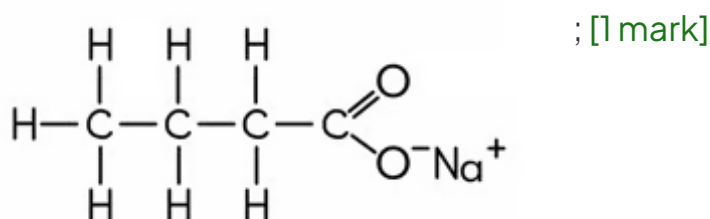
i) The equation for the acid hydrolysis of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ is:



- Correct reactants including balancing numbers; [1 mark]
- Correct products; [1 mark]

ii) The displayed formula of the intermediate formed when $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ is hydrolysed by sodium hydroxide is:

-



[Total: 3 marks]

- Hydrolysis of nitriles can occur either by dilute acid or dilute alkali followed by acidification
- A carboxylic acid is the product in both cases but when hydrolysed by dilute alkali, a carboxylate salt is an intermediate product which requires acidification to turn into the carboxylic acid:
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{NaOH} + 2\text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{COO}^-\text{Na}^+ + \text{NH}_3$
- Acidification of sodium butanoate produces the carboxylic acid $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
- Unlike the formation of nitriles which adds a carbon atom to the carbon chain, hydrolysis does not change the number of carbon atoms

3a

a) The IUPAC name of this hydroxynitrile is:

- 2-hydroxybutanenitrile; [1 mark]

[Total: 1 mark]

- The longest carbon chain is four carbons long, and there is a nitrile group on the first carbon, which means the name must contain ...butanenitrile
- There is a hydroxy group on the second carbon, therefore it has the prefix 2-hydroxy
- This gives the full name 2-hydroxybutanenitrile
- Careful:** It is a common mistake to overlook the carbon of the nitrile group, so giving the name containing ...propanenitrile

3b

acidification

- A carboxylic acid is the product in both cases but when hydrolysed by dilute alkali, a carboxylate salt is an intermediate product which requires acidification to turn into the carboxylic acid:
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{NaOH} + 2\text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{COO}^-\text{Na}^+ + \text{NH}_3$
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3a

a) The IUPAC name of this hydroxynitrile is:

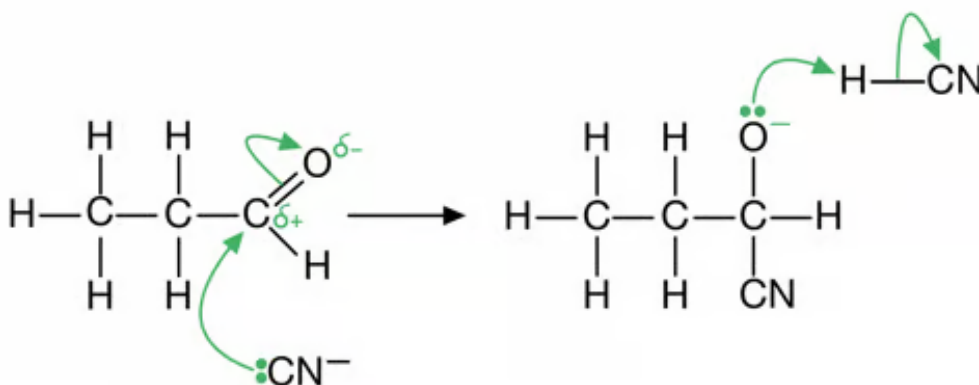
- 2-hydroxybutanenitrile; [1 mark]

[Total: 1 mark]

- The longest carbon chain is four carbons long, and there is a nitrile group on the first carbon, which means the name must contain ...butanenitrile
- There is a hydroxy group on the second carbon, therefore it has the prefix 2-hydroxy
- This gives the full name 2-hydroxybutanenitrile
- **Careful:** It is a common mistake to overlook the carbon of the nitrile group, so giving the name containing ...propanenitrile

3b

b) The completed reaction mechanism shown in Fig. 3.2 for the reaction between propanal and the cyanide ion, CN^- is:



- Lone pair of electrons on the C of CN^- **AND** a curly arrow from the lone pair to δ^+ C atom; [1 mark]
- Curly arrow from $\text{C}=\text{O}$ bond to δ^- O atom; [1 mark]
- Lone pair **AND** negative charge shown on O atom of intermediate **AND** curly arrow from lone pair to H of HCN; [1 mark]
- Curly arrow from $\text{H}-\text{CN}$ bond to C atom; [1 mark]

[Total: 4 marks]

- You must show the dipole on the $\text{C}=\text{O}$ bond by adding in the partial charges
- This means that the $:\text{CN}^-$ ion, which is a nucleophile, will attack the δ^+ C atom, forming a dative covalent bond
- The $\text{C}=\text{O}$ will break by heterolytic fission and a negatively charged intermediate forms
- The negatively charged oxygen atom in the reactive intermediate quickly reacts with aqueous H^+
- In this reaction mechanism, the H^+ comes from HCN, but it could also come from dilute acid or water present in the reaction mixture
 - Acid is present as H_2SO_4 and KCN are used to provide HCN in the reaction
- **Remember:** Curly arrows show the transfer of pairs of electrons and should either start from a lone pair or from a bond

3c

c) The type of reaction mechanism shown in Fig. 3.2 is:

- Nucleophilic addition; [1 mark]

[Total: 1 mark]

- The lone pair of electrons on the $:\text{CN}^-$ ion is attracted to the partially positive carbon atom of the $\text{C}=\text{O}$ bond acting as a **nucleophile**
- Aldehydes undergo addition reactions because they are saturated and contain a $\text{C}=\text{O}$ bond which can break, allowing another species to be **added** to it